

Statistical Comparison: MFE (Optimized) vs Monolith
Welch's t-test ($\alpha = 0.05$, two-tailed). All pages combined.

Condition	Metric	MFE (ms)	Mono (ms)	$\Delta\%$	t-stat	p-value	Sig?	Winner
Warm / No Throttle	SI	2930.4	2900.7	+1.0%	1.03	0.3052	No	—
Warm / No Throttle	FCP	1013.6	947.7	+6.9%	11.98	< 0.001	Yes	Monolith
Warm / No Throttle	LCP	2364.3	2336.1	+1.2%	1.41	0.1607	No	—
Warm / No Throttle	TTI	1620.0	947.7	+70.9%	48.47	< 0.001	Yes	Monolith
Warm / No Throttle	TBT	0.0	0.0	+0.0%	0.00	1.0000	No	—
Cold / No Throttle	SI	2930.6	2899.1	+1.1%	0.81	0.4208	No	—
Cold / No Throttle	FCP	1016.5	947.7	+7.3%	8.80	< 0.001	Yes	Monolith
Cold / No Throttle	LCP	2381.8	2336.5	+1.9%	1.60	0.1106	No	—
Cold / No Throttle	TTI	1665.7	954.5	+74.5%	39.43	< 0.001	Yes	Monolith
Cold / No Throttle	TBT	1.3	0.2	+438.1%	1.52	0.1310	No	—
Warm / 4G	SI	6846.5	7086.7	-3.4%	-3.28	0.0012	Yes	MFE
Warm / 4G	FCP	1988.1	2007.2	-1.0%	-3.94	< 0.001	Yes	MFE
Warm / 4G	LCP	5157.0	4982.1	+3.5%	3.19	0.0015	Yes	Monolith
Warm / 4G	TTI	3477.9	2984.6	+16.5%	4.82	< 0.001	Yes	Monolith
Warm / 4G	TBT	12.2	18.1	-32.9%	-2.64	0.0088	Yes	MFE
Cold / 4G	SI	7117.9	7131.4	-0.2%	-0.21	0.8355	No	—
Cold / 4G	FCP	1991.0	2007.4	-0.8%	-2.34	0.0208	Yes	MFE
Cold / 4G	LCP	5225.4	4983.0	+4.9%	3.62	< 0.001	Yes	Monolith
Cold / 4G	TTI	3571.1	3056.4	+16.8%	3.48	< 0.001	Yes	Monolith
Cold / 4G	TBT	36.1	32.6	+10.7%	0.28	0.7774	No	—
Warm / 3G	SI	12786.8	13311.5	-3.9%	-4.20	< 0.001	Yes	MFE
Warm / 3G	FCP	3341.9	3373.3	-0.9%	-5.95	< 0.001	Yes	MFE
Warm / 3G	LCP	9320.1	8676.6	+7.4%	4.88	< 0.001	Yes	Monolith
Warm / 3G	TTI	5934.5	4937.3	+20.2%	5.83	< 0.001	Yes	Monolith
Warm / 3G	TBT	11.9	15.8	-24.7%	-1.82	0.0691	No	—
Cold / 3G	SI	13045.0	13246.2	-1.5%	-2.04	0.0439	Yes	MFE
Cold / 3G	FCP	3345.5	3373.2	-0.8%	-3.53	< 0.001	Yes	MFE
Cold / 3G	LCP	9442.8	8671.4	+8.9%	4.35	< 0.001	Yes	Monolith
Cold / 3G	TTI	6134.5	5069.6	+21.0%	4.41	< 0.001	Yes	Monolith
Cold / 3G	TBT	50.1	25.4	+97.0%	1.35	0.1797	No	—